

Mohammed Owda

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EDUCATION & DISTINCTIONS

Western University

BESc, Software Engineering — CGPA: 82%

London, ON

Sept 2023 – Apr 2027

- **Dean's Honour List** (2024–2025)
- **RBC Design Thinking Program** (Fall 2025, \$3,500 stipend)
- **1st Place:** Automotive Innovation Challenge, Hack Western, MSAHacks
- **Chorley & Bisset Ltd. Consulting Engineers Award** (2025), \$2,800

RESEARCH EXPERIENCE

Smart Mobility Research Lab, Western University

Undergraduate Research Assistant

London, ON

May 2025 – Present

- Developed a **multi-view AI dashboard** with real-time YOLOv11 overlays and event logging.
- Engineered a **ROS 2 pipeline** (rclpy) for multi-camera discovery and modular detection publishing.

SELECTED PROJECTS

Autonomous Thermal Object Detection

YOLOv11, NCNN (ARM), Raspberry Pi, Federated Learning

Mar 2025 – Apr 2025

- **Optimized on-device YOLO** on Raspberry Pi to **0.625s/image** via quantization/pruning and runtime tuning.

QCare: ER Virtual Waiting Room

React, Flask, SQLAlchemy, OpenAI

Feb 2025 – Mar 2025

- Built an **AI-assisted triage** and queuing system with real-time clinician dashboard.

Vinculum (Collaborative Web App)

React, Node.js, WebSockets, WebRTC

Nov 2024 – Dec 2024

- Enabled synchronized **multi-user interaction with voice chat** in a shared 2D canvas.

E-Commerce Platform for Western Baja

React, Node.js, PostgreSQL

Sept 2024 – Oct 2024

- Built **modular frontend** and **REST APIs** for parts registration, order flows, and admin controls.

LEADERSHIP & SERVICE (SELECTED)

Western Engineering Competition (WEC)

Vice President, Technical Operations

London, ON

Mar 2025 – Present

- Delivered digital infrastructure for **650+ registrants**; managed website/LMS operations.

Engineers Without Borders (Western Chapter)

Vice President of Projects

London, ON

Sept 2023 – Present

- Led STEM workshops for **200+ youth**; co-organized **Sustainability Hackathon** with **100+ participants**.

Western Foot Patrol

Operations Manager

London, ON

Sept 2023 – Present

- Coordinated volunteer scheduling and logistics to maintain campus safety services.

OTHER EXPERIENCE

DECO Repair

Sales Representative

London, ON

Apr 2024 – Sept 2024

- Generated **\$13,000+** in sales through consultative engagement; improved client communication.

TECHNICAL SKILLS

Programming Languages: Python; C/C++; Java; JavaScript/TypeScript; SQL; Bash; HTML/CSS

Web & APIs: React, Next.js, Node.js/Express, FastAPI, REST, WebSockets, SDLC (Agile, Waterfall)

Robotics & CV: YOLO (v8/v11), OpenCV, ROS 2 (rclpy), NCNN/TensorRT (basics)

Cloud/DevOps: Docker, Git/GitHub, AWS (EC2/S3/IAM), Pytest, UML, basic CI/CD, MS Office